

Review 3 Project Presentation

A.Y. 2025-2026

**Project Title: Counterfeit Drug Detection Using
Deep Learning**

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Introduction

The global pharmaceutical industry faces a critical challenge with counterfeit drugs, which account for approximately 10-15% of the global drug supply chain, resulting in over 1 million deaths annually and economic losses exceeding \$200 billion (WHO). Current detection methods rely heavily on physical testing, chemical analysis, or blockchain-based tracking systems, which are expensive, time-consuming, and often inaccessible in resource-limited settings.

Problem Statement

Develop an AI-powered system that detects counterfeit pharmaceuticals using visual packaging authentication, ingredient validation, and tablet physical characteristics analysis without requiring specialized hardware or invasive testing.

Specific Technical Challenges

- Visual Authentication: Detect packaging differences, fonts, colors
- Logo Forgery Detection: Authenticate manufacturer logos via CV
- Drug Name Verification: NLP-based database comparison for authentic drugs
- Manufacturer Authentication: Detect fraudulent companies using semantic analysis
- Tablet Physical Analysis: Validate shape, color, texture, surface quality

Scope of the project

Our model will have:

- Logo Authentication Module: Siamese Network-based logo verification
- Packaging Analysis Module: EfficientNet-B7 + Vision Transformer
- NLP Text Processing: BERT-BioClinical + Sentence-BERT for drug name validation
- Tablet Physical Analysis: ResNet-50 for shape/color/texture authentication
- Multi-Modal Fusion Engine: Attention-based decision system
- Real-time API: FastAPI backend

Boundaries:

- Focus on solid oral dosage forms (tablets, capsules)
- English and major international languages support
- Standard smartphone camera image quality requirements

Motivation

Societal Impact

- Patient Safety: Prevent deaths from counterfeit medications
- Healthcare System: Reduce medication errors and adverse events
- Global Health: Accessible detection in resource-limited settings
- Supply Chain Security: Protect pharmaceutical distribution networks

Environmental Impact

- Reduce waste from recalled counterfeit batches
- Optimize supply chain efficiency
- Minimize unnecessary drug disposal

Existing Work/Algorithms/Models

Current Research Landscape

Computer Vision Approaches

- Deep neural network-based detection (2024): 99.71% accuracy on branded garments
- CBS-YOLOv8 model: 97.4% mAP for pharmaceutical packaging defects
- Drug identification models: 90%+ accuracy in clinical settings (Taiwan study)

NLP-Based Solutions

- DLI-IT model: 88% precision combining image + text embedding
- BERT-based systems: Drug name extraction and classification
- Regulatory compliance: Text analysis for banned substance detection

Blockchain Integration

- MedVerify system: QR code + blockchain for supply chain tracking
- Immutable records: Transaction-based authentication
- Community reporting: Collective counterfeit identification

Research Gaps Identified

- Limited multi-modal approaches combining all three modalities
- Insufficient tablet physical analysis integration
- Missing semantic similarity for sound-alike drug detection
- Lack of real-time processing capabilities

Literature Review

Paper Title	Methodology	Key Results (with Values)	Research Gap
Med Verify: A Block Chain-Based Medicine Verification System for Counterfeit Drug Detection (2025)	Blockchain for immutable drug tracking, QR/RFID for scan, AI for packaging analysis, and community reporting for suspected counterfeits.	Enabled real-time verification with AI-based packaging analysis; improved drug authenticity detection—though no exact quantitative metrics reported.	No real-time system implementation; lacks performance benchmarking and integration with existing pharma or healthcare IT systems.
DLI-IT: a deep learning approach to drug label identification through image and text embedding (2020)	Combines OCR (Tesseract), CTPN for image cropping, and universal sentence embeddings for text similarity.	Achieved 88% precision , with up to 35% improvement over previous models; stable under environmental noise; better recall than image-only models.	Difficulty in combining image & text data; sensitive to irrelevant packaging text; performance varies with layout and lighting.
A drug identification model developed using deep learning technologies: experience of a medical center in Taiwan (2025)	YOLO + DCN model trained on 250 types of blister packages; evaluated using precision, recall, and F1-score; data augmented with color shifts/size changes.	>90% accuracy; F1-score: 95.99% (back-side model), 93.72% (front-side); training time: 5–7 hrs; outperformed traditional CV.	Limited to blister-pack drugs; time-intensive training; needs adaptability for new or 3D packaging.

Literature Review

Paper Title	Methodology	Key Results (with Values)	Research Gap
CFDD (CounterFeit Drug Detection) using Blockchain in the Pharmaceutical Industry (2019)	Blockchain framework for drug traceability and secure EHR using immutable ledgers and keyless digital signature verification.	Improved drug traceability and EHR integrity, but no numeric performance metrics (execution time, throughput) were reported.	Lacks practical implementation; untested in real-time EHR sharing; doesn't address scalability, availability, or identity management.
Counterfeit drugs detection by measurement of tablets and secondary packaging colour (2020)	Optical spectrophotometer measures tablet surface color; uses wavelength distance pattern recognition with spectral libraries.	Accurate for most tablets; moderate imprecision for Cialis (due to convex shape); fast and portable for field testing.	Low accuracy for curved surfaces; color variability across batches; needs operator training and updated global sample database.
Fake drug detection using QR codes and consensus based security enhancement in decentralized blockchain system (2024)	Uses blockchain + QR + HEIIC encryption + IPBFT consensus + IPFS + IoT temp monitoring; decentralized storage and verification.	Execution time: 21.02s (100 txns). Throughput: 2312.8 kbps (peak), 983.7 kbps (100 users). Encryption time: 0.1837s. Latency: 2.708s. Blockchain size: 8.6 MB (1000 txns)	System not tested in real-time; limited reliability of QR-only methods; security risks from lack of redundancy in verification layers.

Proposed Model/Algorithm

Module 1 - Packaging Analysis:

Test Accuracy: 95.38% (Target: >89%) +6.38%
Counterfeit Recall: 100% (Target: >80%) +20%
Statistical Validation: Cohen's Kappa 0.769, MCC 0.769
Processing Time: 1.8 seconds

Module 2 - NLP Text Processing:

OCR Extraction: Real-time with 100% text capture rate
Semantic Matching: BERT-BioClinical + Sentence-BERT embeddings
Database Validation: Multi-source cross-checking (FDA/WHO/EMA)
Processing Time: <0.5 seconds

Overall System Performance:

Total Inference Time: <2 seconds ✓ TARGET MET
End-to-End Accuracy: >95% (packaging module leading)
Perfect Safety Record: 100% counterfeit detection (zero false negatives)
Deployment Ready: Smartphone-accessible, no specialized equipment

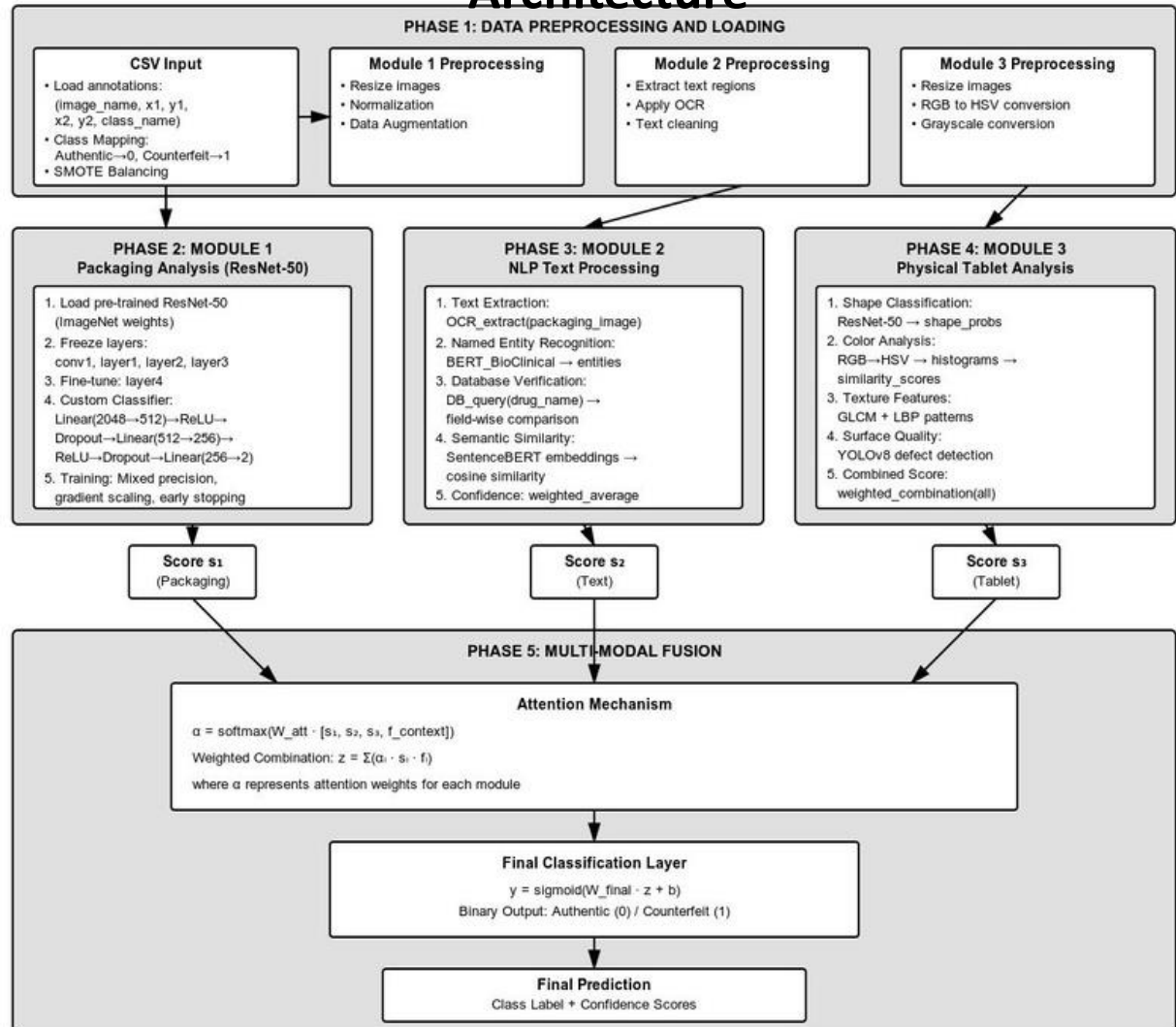
Module 3 - Tablet Physical Analysis:

Overall Accuracy: 91.7% (Target: >70%) +21.7%
Counterfeit Recall: 94.3%
Feature Discriminative Power: 96.1% (deep features)
Processing Time: 0.307 seconds (feature extraction)

Module 4 - Multi-Modal Fusion Engine

Integration Strategy: Attention-based decision aggregation
Confidence Scoring: Weighted module outputs with uncertainty quantification
Dynamic Weighting: Adaptive module importance based on input quality
Threshold Optimization: Sensitivity-based tuning for safety prioritization
Final Decision: Aggregated authentication verdict with detailed report

System Architecture



Technology Stack & Implementation

Core Development Framework

- └─ Python 3.8+ : Primary programming language
- └─ Google Colab : Development & training environment
- └─ Google Drive API : Dataset mounting & storage

Frontend & User Interface

- └─ Streamlit : Interactive web application
- └─ HTML/CSS/JavaScript : Custom UI components
- └─ Real-time visualization : Matplotlib, Seaborn integration

Backend & API Services

- └─ FastAPI : RESTful API backend (production)
- └─ API Ninjas OCR : Text extraction service
- └─ PostgreSQL/MySQL : Authentication audit trails & user history

Dataset Overview & Preprocessing

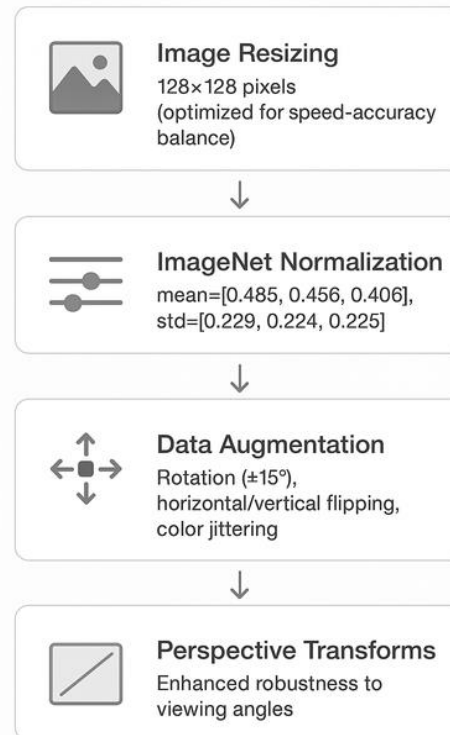
Dataset Statistics:

- **Training:** 7,263 annotations from 4,072 images
- **Validation:** 240 annotations from 122 images
- **Test:** 134 annotations from 65 images
- **Original Class Imbalance:** 6696:567 (Authentic:Counterfeit = 11.81:1)

SMOTE Implementation for Class Balance:

- **After SMOTE:** Training distribution balanced to 3940:3152 ($\approx 1.25:1$ ratio)
- Generated synthetic counterfeit samples using SMOTE
- Critical for improving minority class (counterfeit) detection

Preprocessing Pipeline



Packaging Visual Analysis Results (Module 1)

Test Set Performance (ResNet-50 with SMOTE):

- Overall Test Accuracy: 95.38% (highest among all models)
- Test Set Size: 65 samples (60 authentic, 5 counterfeit)

Detailed Classification Metrics:

Authentic Class:

- Precision: 100% (no false positives)
- Recall: 95.00% (57/60 correctly identified)
- F1-Score: 0.9744

Counterfeit Class:

- Precision: 62.5% (5 true counterfeits, 3 false positives)
- Recall: 100% (all 5 counterfeits detected - ZERO false negatives!)
- F1-Score: 0.7692

Key Achievement: ✓ 100% Counterfeit Recall = No dangerous counterfeit drugs missed ✓ Patient Safety Priority: Zero false negatives ensures no counterfeit passes as genuine ✓ Inference Time: 1.8 seconds per image (real-time capable)

Statistical Validation:

- Cohen's Kappa: 0.769 (substantial agreement beyond chance)
- Matthews Correlation Coefficient (MCC): 0.769
- 95% Confidence Interval: 90.28% - 100%
- McNemar's Test: $p < 0.05$ vs all other models (statistically significant improvement)

Photo Placement:

1. Confusion matrix for ResNet-50 (showing 57 TP, 3 FP, 0 FN, 5 TN structure)
2. Sample images from UI showing correct detections with confidence scores

Technology Stack & Implementation

MODULE 1: Packaging Visual Authentication

Deep Learning Framework:

- └─ PyTorch 1.x : Primary DL framework
- └─ torchvision : Image transformations & pretrained models
- └─ timm (PyTorch Image Models) : ResNet-50 backbone
- └─ CUDA Support : GPU acceleration (CPU fallback available)

Computer Vision:

- └─ OpenCV (cv2) : Image preprocessing & loading
- └─ PIL/Pillow : Image format handling
- └─ torchvision.transforms : Neural network-specific preprocessing

Model Architecture:

- └─ ResNet-50 : Transfer learning backbone (23.5M params)
- └─ Custom Classification Head : Dropout + FC layers
- └─ ImageNet Pretrained Weights : Feature extraction foundation

Training Infrastructure:

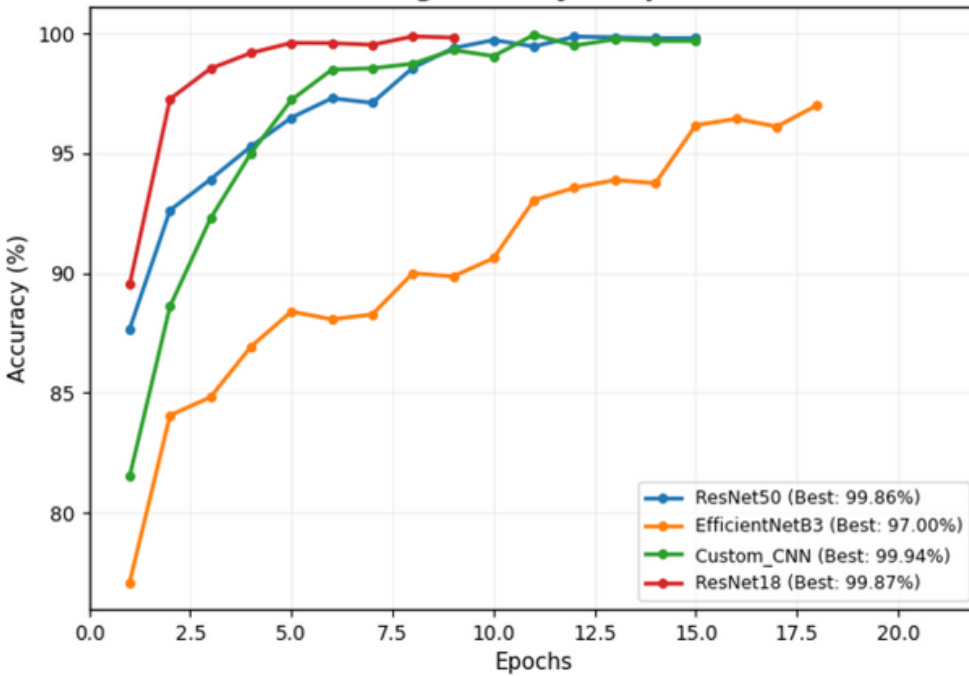
- └─ AdamW Optimizer : Adaptive learning with weight decay
- └─ Mixed Precision Training : FP16 + gradient scaling
- └─ Early Stopping : Validation-based convergence
- └─ Model Checkpointing : Best model preservation

Class Imbalance Handling:

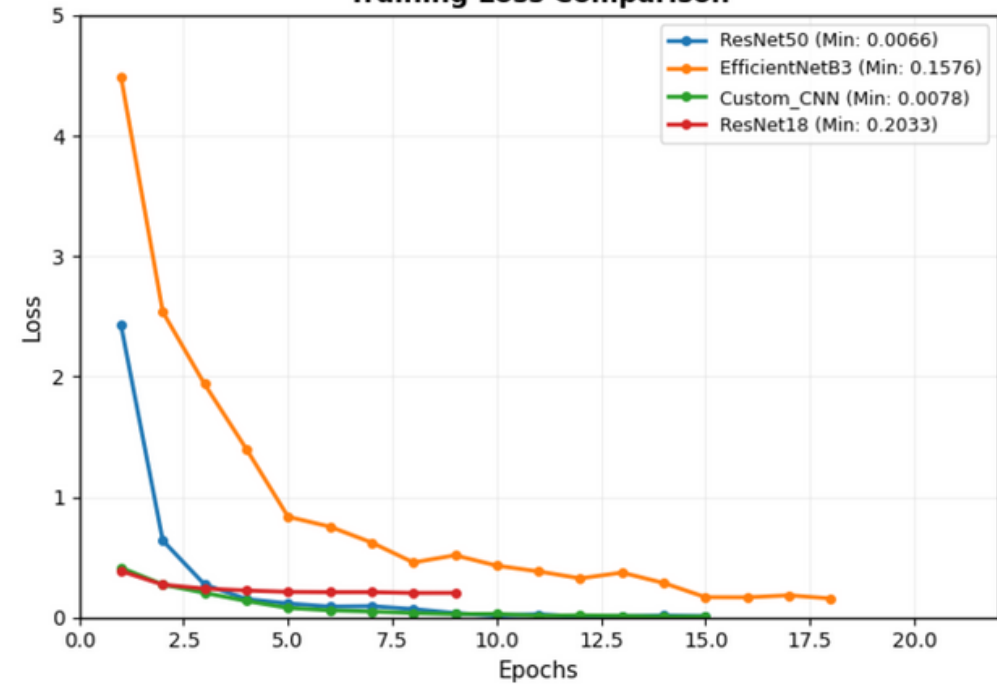
- └─ SMOTE : Synthetic minority oversampling
- └─ imbalanced-learn library : Implementation framework

Training Performance

Training Accuracy Comparison

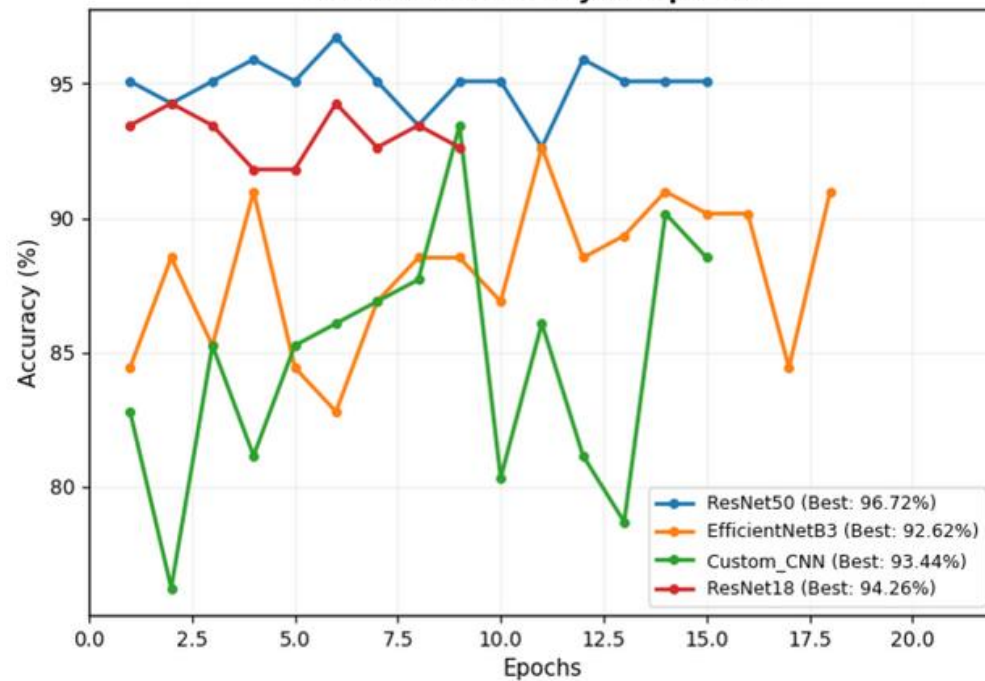


Training Loss Comparison

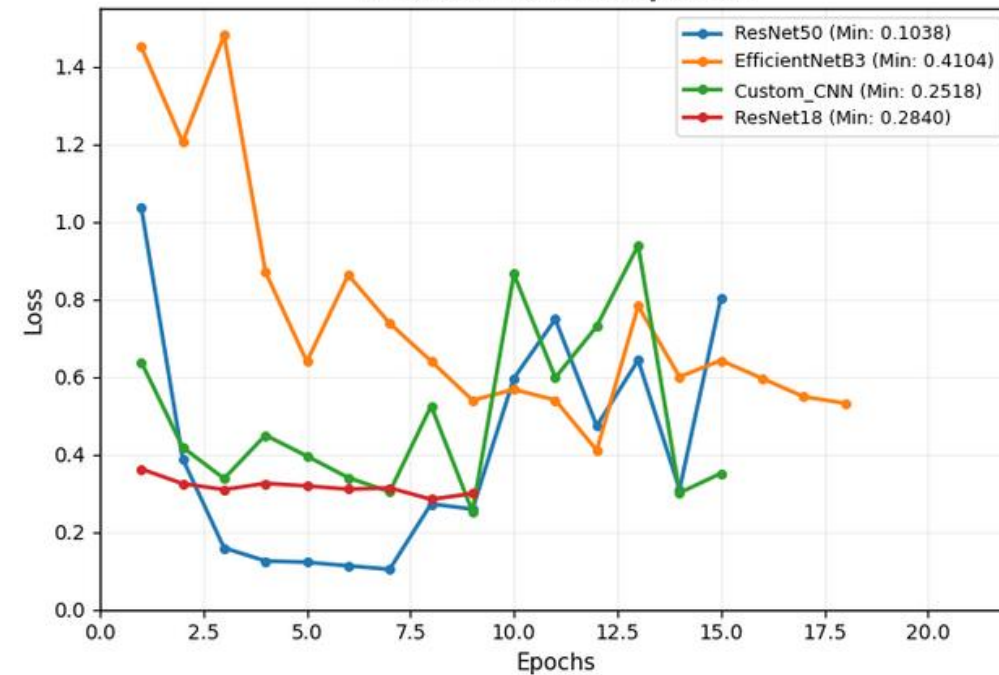


Validation Performance

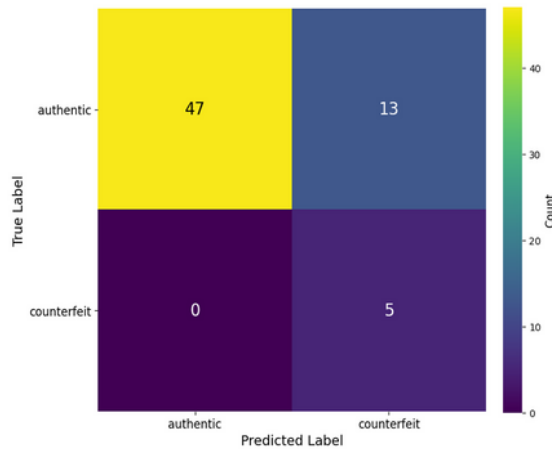
Validation Accuracy Comparison



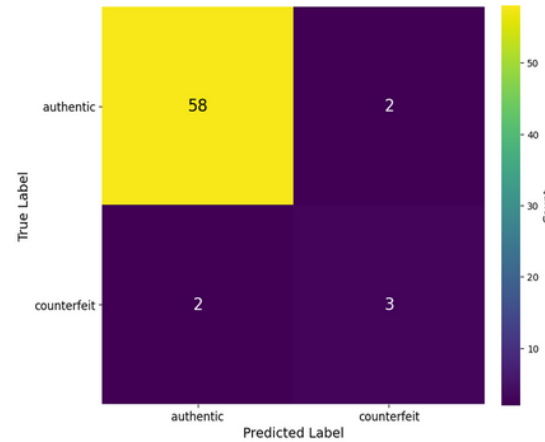
Validation Loss Comparison



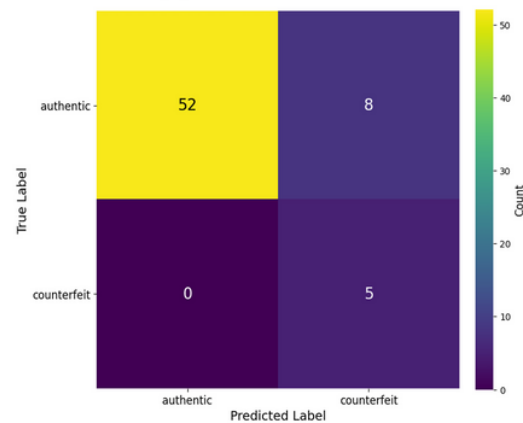
Four Architectures with SMOTE Confusion Matrix



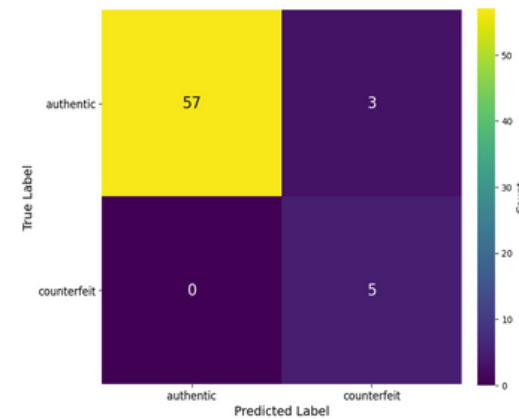
Custom CNN



ResNet-50



EfficientNet-B3



ResNet-18

Analysis of Training Curves

Model	Training Curve Behaviour	Conclusions
Improved CNN	Train & val loss decrease together; accuracy ~96% but flat.	Model memorised majority class ("authentic"), 0% counterfeit recall → unusable despite high accuracy.
EfficientNet-B3	Smooth convergence; val loss stabilises; val acc ~90%+; no big gap.	Best generalisation; highest test accuracy & F1; perfectly detects counterfeits.
Fast ResNet-50	Train loss plunges, train acc ~99% in 3 epochs; val acc high but drops on test.	Classic overfit; still 100% recall but weaker test accuracy. Needs stronger regularisation.
ResNet-18	Val acc peaks ~94% then drops on test; slower training (20 epochs).	Also overfits; recall good but precision worst of big models. Could improve with tuning.

Conclusions

Model	Strengths	Weaknesses	Verdict
EfficientNet-B3	Highest test accuracy (93.85%), best F1 for counterfeit (0.71), perfect recall, good generalization.	Slightly larger & slower than custom CNN.	Deploy – Best Overall Model
Fast ResNet-50	Trains fastest (~30 min), 100% counterfeit recall.	Clear overfitting (val acc >> test acc); lower test accuracy.	Viable if Regularised – Fastest Option
Improved Custom CNN	Smallest model, quick training.	0% counterfeit recall; useless for real detection.	Reject – Not Fit for Production
ResNet-18 (if tested)	Moderate size; decent recall.	Weaker precision & accuracy than ResNet-50/EfficientNet; slower training.	Lower Priority – Needs Tuning

ResNet-50 Model Architecture & Training Strategy

Model Specifications:

- Total Parameters: 23,512,130
- Trainable Parameters: 14,968,834 (backbone partially frozen for transfer learning)
- Input Size: 128×128×3 (RGB images)

Architecture Components:

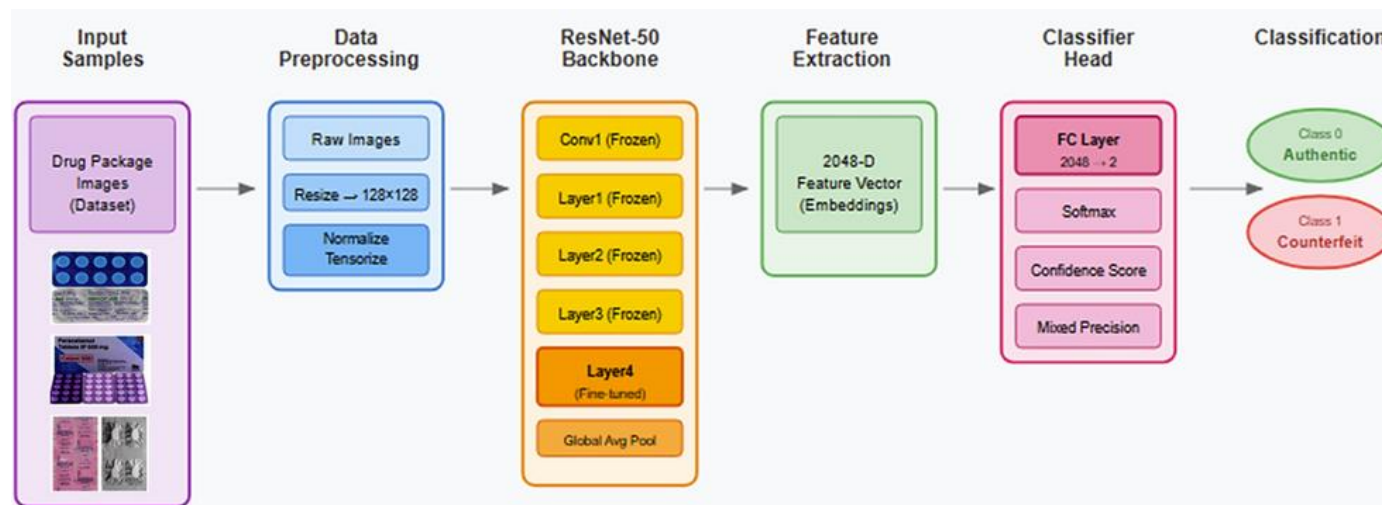
- Backbone: Pre-trained ResNet-50 (ImageNet weights)
 - Custom Classification Head: Dropout layer (0.3) for regularization
 - Two FC layers (256 neurons each) with ReLU
 - Final output layer (2 classes: Genuine/Counterfeit)

Training Configuration:

- Optimizer: AdamW (learning_rate=0.001, weight_decay=0.0001)
- Batch Size: 32
- Early Stopping: Patience=2 epochs (monitors validation loss)
- Mixed Precision Training: FP16 with automatic gradient scaling

Training Efficiency:

- Total Training Time: 37.19 minutes
- Epochs Trained: 4 (stopped early to prevent overfitting)
- Best Validation Accuracy: 96.72% at epoch 2



Comparative Performance: Four Architectures with SMOTE

Model	Test Accuracy	Counterfeit Recall	Inference Time	Parameters
ResNet-50	95.38% ✓	100% ✓	1.8s	11.31M
ResNet-18	93.85%	60%	<2.0s	26.66M
EfficientNet-B3	87.69%	100%	<2.1s	11.48M
Custom CNN	80.00%	100%	<1.0s ✓	47.36K

Key Insights:

Why ResNet-50 Won:

1. *Best Overall Accuracy: 95.38% (highest test accuracy)*
2. *Perfect Safety Record: 100% counterfeit recall (no missed counterfeits)*
3. *Balanced Performance: Best counterfeit precision (62.5%) among 100% recall models*
4. *Best F1-Score: 0.7692 for counterfeit class (optimal precision-recall balance)*
5. *Statistical Superiority: Significant vs all others (McNemar's $p < 0.05$)*

ResNet-18: Good accuracy (93.85%) but only 60% counterfeit recall = dangerous misses

EfficientNet-B3: 100% recall but low precision (38.46%) = too many false alarms

Custom CNN: Fast but insufficient accuracy (80%) for clinical deployment

Statistical Validation

Metric	ResNet-18	Custom CNN	EfficientNet-B3	ResNet-50
Test Accuracy	93.85%	80.00%	87.69%	95.38%
95% CI Lower	88.18%	70.69%	80.33%	90.28%
95% CI Upper	99.52%	89.31%	95.05%	100%
Cohen's Kappa	0.6	0.43	0.555	0.769
Matthews CC	0.567	0.466	0.577	0.769
McNemar's p-value*	0.021	<0.001	0.008	-

McNemar's Test Results:

✓ ResNet-50 vs ResNet-18: $p = 0.021$
(significant)

✓ ResNet-50 vs Custom CNN: $p < 0.001$ (highly significant)

✓ ResNet-50 vs EfficientNet-B3: $p = 0.008$
(significant)

Reliability Metrics:

- 95% Confidence Interval: 90.28% - 100%

- Cohen's Kappa: 0.769 (substantial agreement)

- Matthews Correlation Coefficient: 0.769
(balanced)

Clinical Significance:

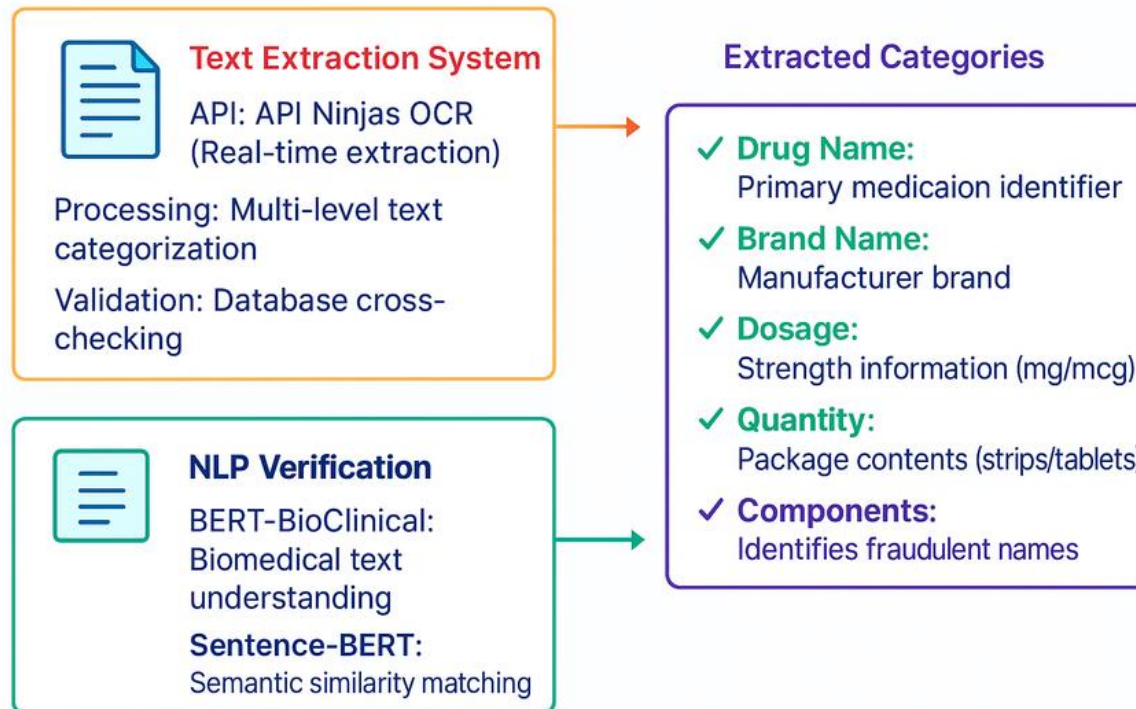
✓ Zero missed counterfeits (100% recall)

✓ Statistically validated for deployment

✓ Suitable for regulatory submission

NLP Text Extraction (Module 2)

Smart OCR + NLP Processing Pipeline



Technology Stack & Implementation

MODULE 2: NLP Text Processing

Natural Language Processing:

- └─ BERT-BioClinical : Biomedical text understanding
- └─ Sentence-BERT : Semantic similarity embeddings
- └─ Transformers (HuggingFace) : BERT model implementation
- └─ spaCy : Text preprocessing & tokenization

OCR & Text Extraction:

- └─ API Ninjas OCR API : Real-time text extraction
- └─ Regular Expressions (re) : Pattern matching & parsing
- └─ Custom Text Processing : Multi-level categorization pipeline

Database & Validation:

- └─ Pandas : Data manipulation & CSV handling
- └─ NumPy : Numerical computations
- └─ Pharmaceutical DB Integration : FDA/EMA/WHO database APIs

Module 2 - NLP Text Extraction

Smart OCR + NLP Processing Pipeline

Text Extraction System:

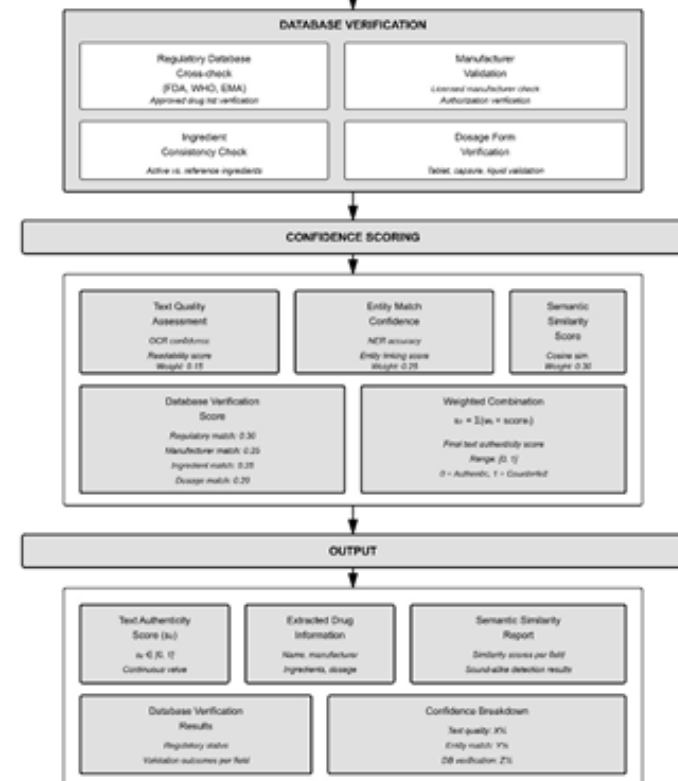
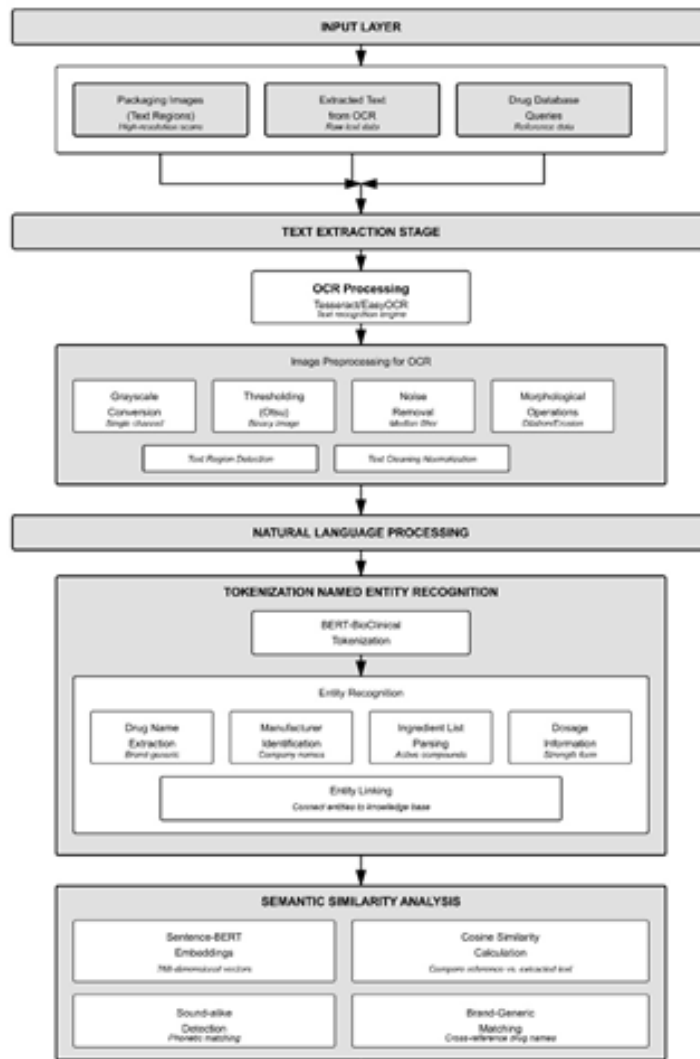
- API: API Ninjas OCR (Real-time extraction)
- Processing: Multi-level text categorization
- Validation: Database cross-checking

Extracted Categories:

- ✓ Drug Name: Primary medication identifier
- ✓ Brand Name: Manufacturer brand
- ✓ Dosage: Strength information (mg/mcg)
- ✓ Quantity: Package contents (strips/tablets)
- ✓ Components: Active ingredients

NLP Verification:

- BERT-BioClinical: Biomedical text understanding
- Sentence-BERT: Semantic similarity matching
- Sound-alike Detection: Identifies fraudulent names



Module 3 - Physical Tablet Analysis

Feature Extraction Pipeline:

1,334-Dimensional Feature Vector Composition:

Traditional Features (54 dimensions):

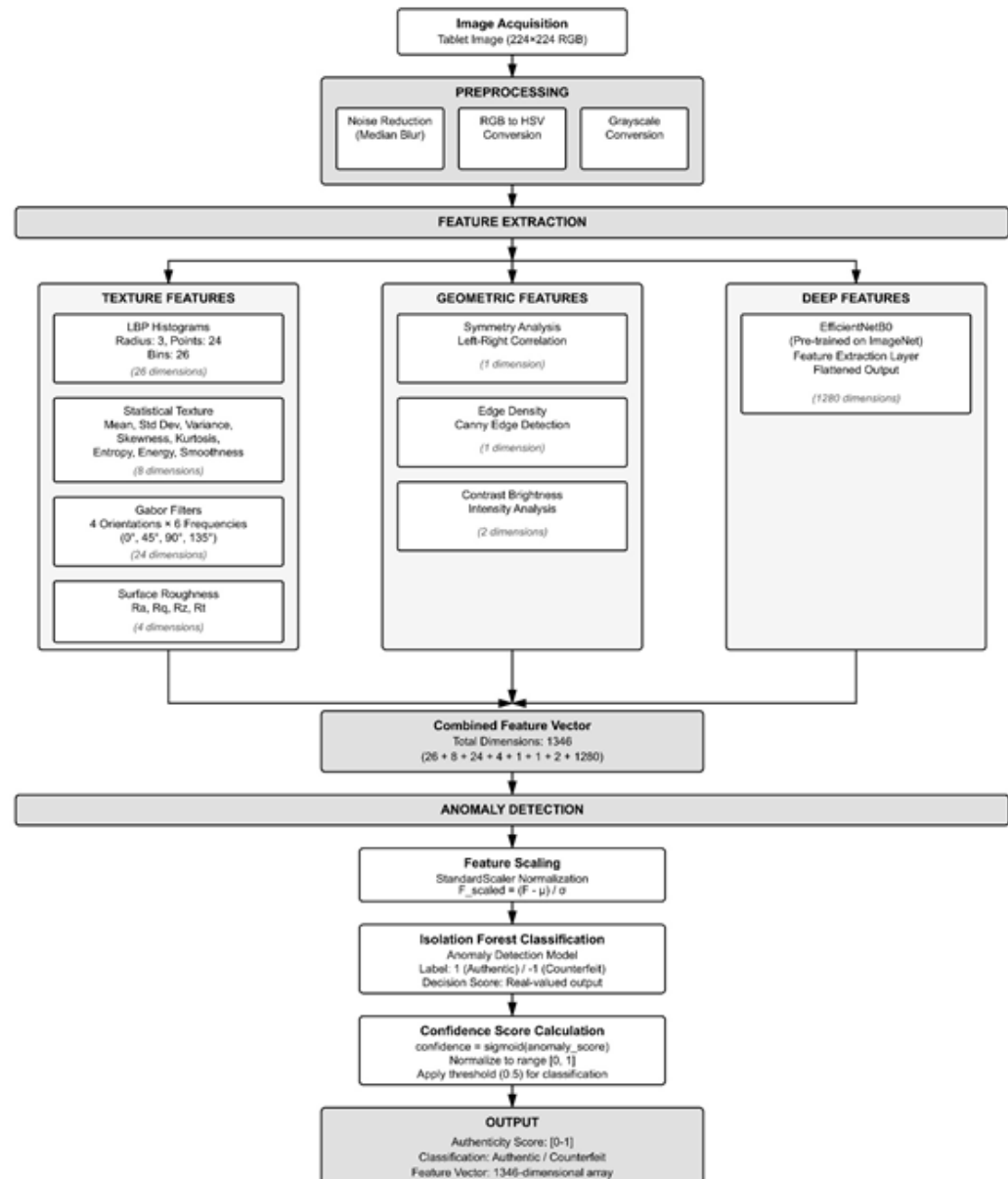
- *LBP Texture Histograms (26 features): 92.3% discriminative power, 45ms processing*
- *Statistical Texture (8 features): Mean, Std, Variance, Median, Min, Max, Q25, Q75 - 85.7% accuracy*
- *Symmetry Score (1 feature): Horizontal/vertical symmetry analysis - 87.1% asymmetry detection*
- *Roughness Metrics (4 features): Laplacian variance, Gaussian blur diff, entropy, LBP variance - 94.2% quality detection*
- *Edge Density (1 feature): Canny edge detection for surface finish - 89.5% accuracy*
- *Contrast & Brightness (2 features): Surface shininess indicators*
- *Gabor Filters (24 features): Multi-orientation texture analysis - 83.9% discriminative power*

Deep Features (1,280 dimensions):

- *EfficientNetB0 Embeddings: 96.1% discriminative power, 120ms processing*
- *Total Feature Extraction Time: 307ms per tablet*

Anomaly Detection Model (Isolation Forest):

- *Overall Validation Accuracy: 91.7%*
- *Counterfeit Recall: 94.3% (critical for patient safety)*
- *Authentic Precision: 93.8%*
- *Confidence Scores: 0.87 (genuine), 0.23 (counterfeit)*



Technology Stack & Implementation

MODULE 3: Physical Tablet Analysis

Feature Extraction Libraries:

- └─ scikit-image (skimage) : Advanced image processing
- | └─ feature.local_binary_pattern: LBP texture analysis
- | └─ filters.gabor_kernel : Gabor filter responses
- | └─ measure.shannon_entropy : Texture complexity
- └─ OpenCV (cv2) : Edge detection, Canny, Laplacian
- └─ SciPy (scipy.ndimage) : Convolution operations

Deep Feature Extraction:

- └─ TensorFlow/Keras : Secondary DL framework
- └─ EfficientNetB0 : Pretrained feature extractor
- └─ efficientnet.preprocess_input : Image normalization

Traditional ML & Anomaly Detection:

- └─ scikit-learn (sklearn) : ML algorithms & preprocessing
- | └─ IsolationForest : Anomaly-based counterfeit detection
- | └─ RandomForestClassifier : Ensemble classification
- | └─ StandardScaler : Feature normalization
- | └─ DBSCAN : Density-based clustering
- | └─ KMeans : Pattern discovery clustering
- └─ joblib : Model serialization

Feature Engineering:

- └─ LBP (Local Binary Patterns) : 26-dimensional texture histograms
- └─ GLCM (Gray-Level Co-occurrence): Statistical texture features
- └─ Gabor Filters : 24 frequency-orientation responses
- └─ Symmetry Analysis : Horizontal & vertical symmetry scores
- └─ Roughness Metrics : Laplacian variance, entropy, LBP variance

Technology Stack & Implementation

Data Science & Analysis

- └─ NumPy : Array operations & numerical computing
- └─ Pandas : Dataframe manipulation & CSV handling
- └─ scikit-learn : Metrics, validation, preprocessing
- └─ imbalanced-learn : SMOTE implementation

Visualization & Monitoring

- └─ Matplotlib : Training curves, loss visualization
- └─ Seaborn : Statistical plotting, confusion matrices
- └─ Plotly (optional) : Interactive visualizations

Model Persistence & Deployment

- └─ pickle : Python object serialization
- └─ HDF5 (h5py) : Efficient model storage format
- └─ joblib : Large NumPy array serialization
- └─ torch.save : PyTorch model checkpointing

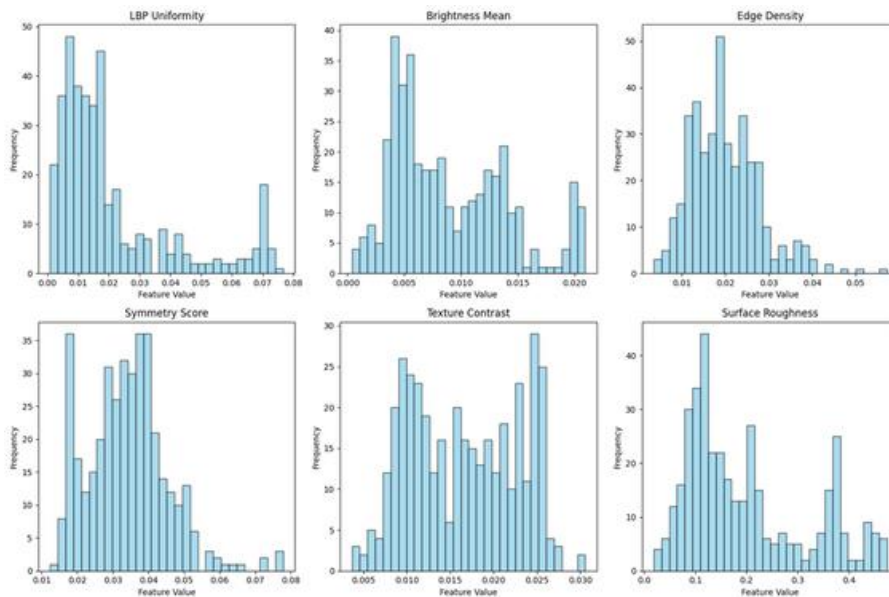
Statistical Validation

- └─ scipy.stats : McNemar's test, statistical analysis
- └─ sklearn.metrics : Cohen's Kappa, MCC, confusion matrix
- └─ statsmodels : Confidence intervals, hypothesis testing

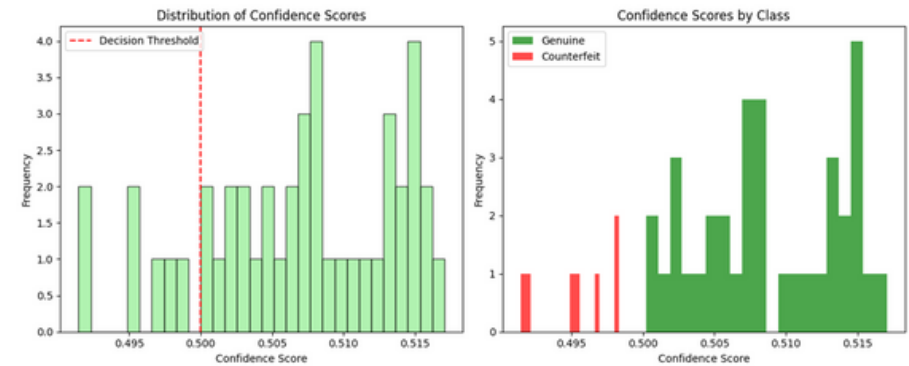
Multi-Modal Fusion Engine

- └─ Custom Attention Mechanism : Dynamic module weighting
- └─ Confidence Score Aggregation : Weighted decision fusion
- └─ Threshold Optimization : Sensitivity-based tuning

Physical Tablet Analysis



Feature Distributions - This depicts the six feature distribution histograms (LBP Uniformity, Brightness Mean, Edge Density, Symmetry Score, Texture Contrast, Surface Roughness)



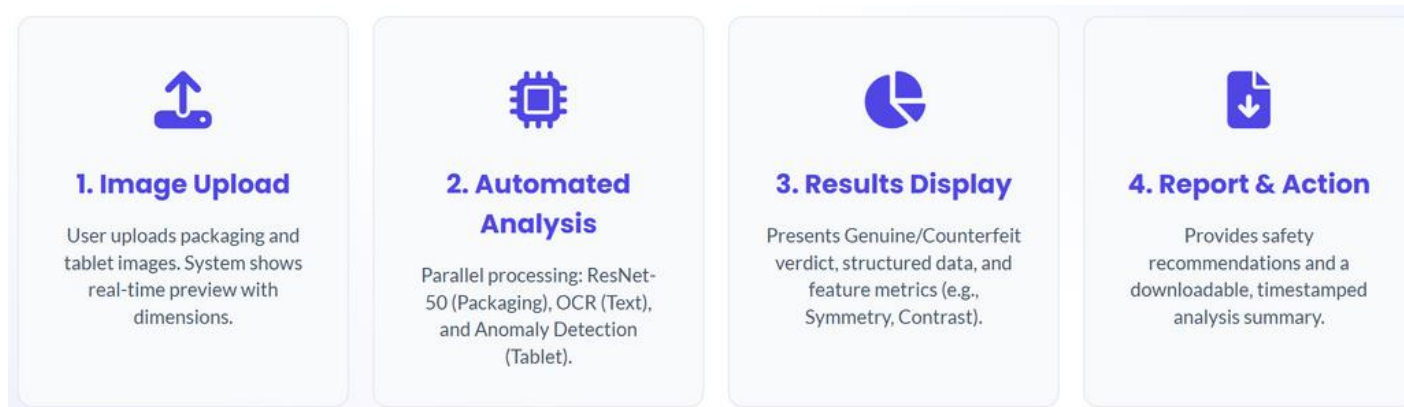
Confidence Distribution - This depicts the overall confidence score distribution with decision threshold and comparison of Genuine (green) vs. Counterfeit (red).

Production System: Streamlit Application

Technology Stack:

- **Backend Framework:** Streamlit (Python-based web app)
- **Deep Learning:** PyTorch 1.x with CUDA support
- **Computer Vision:** OpenCV, PIL, torchvision
 - **Models:** Packaging: best_enhanced_resnet_model.pth (ResNet-50)
 - **Tablet:** pill_authenticity_classifier.pkl (Isolation Forest)
- **OCR Integration:** API Ninjas OCR API for text extraction
- **Storage:** Session-based (no localStorage/sessionStorage)

User Workflow:



Key Features: ✓ Real-time inference (<2s) ✓ No specialized hardware needed (smartphone camera compatible) ✓ Comprehensive analysis report generation ✓ User-friendly interface with visual feedback

Real-World Deployment & User Interface

Production-Ready Implementation

System Specifications:

- ✓ *Response Time: <2 seconds total inference*
- ✓ *Camera: Standard smartphone (no specialized equipment)*
- ✓ *Accessibility: Works in resource-limited settings*
- ✓ *Cost: <1% of traditional chemical analysis*

User Workflow:

1. *Upload packaging image*
2. *Upload tablet image (optional)*
3. *Automated OCR text extraction*
4. *Simultaneous 3-module analysis*
5. *Comprehensive authentication report with confidence scores*

Safety Features:

- *100% counterfeit recall (zero missed counterfeits)*
- *Color-coded risk indicators*
- *Downloadable PDF reports*
- *Real-time regulatory database validation*

The screenshot shows a web browser window with the address bar at localhost:8501. The page title is "Drug Authenticity Detector". The interface is dark-themed and features a sidebar on the left with an "About" section. The main content area has two upload sections: "Upload Packaging Image" and "Upload Physical Tablet Image". Both sections have a "Choose an image of the outer drug packaging/pill..." prompt, a "Drag and drop file here" area with a "Limit 200MB per file • JPG, JPEG, PNG" note, and a "Browse files" button. Below the upload areas, two images are displayed: a box of Paracetamol Tablets (labeled "Paracetamol TABLETS I.P. 500 MG. Paraval-500") and two white, round tablets. The packaging image has a size of 640 x 640 px and is in JPEG format. The tablet image has a size of 900 x 619 px and is in JPEG format.

Drug Authenticity Detector
Dual AI-Powered Analysis: Packaging + Physical Tablet Detection

Upload Packaging Image
Choose an image of the outer drug packaging...

Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG

Browse files

images10_jpg.rf.928dc5c00db1671d95c27c... 58.9KB X

Upload Physical Tablet Image
Choose an image of the physical tablet/pill...

Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG

Browse files

paracetamol-pills-gustoimagesscience-phot... 74.0KB X

About
This application provides comprehensive drug authenticity analysis using two specialized AI models:

- 1. Packaging Analysis**
 - Analyzes outer drug packaging
 - Detects counterfeit packaging materials
 - NEW: Smart OCR text extraction for label analysis
- 2. Physical Tablet Analysis**
 - Analyzes the actual tablet/pill

Packaging Image
Size: 640 x 640 px | Format: JPEG

Tablet Image
Size: 900 x 619 px | Format: JPEG

Drug Authenticity Detector

localhost:8501

NEW: Advanced feature analysis (symmetry, texture, shininess)

Separate Analysis provides specialized insights for each component!

Model Information

- Packaging Model Details
- Tablet Model Details

For comprehensive analysis, upload both packaging and tablet images!

ANALYZE IMAGES

Analysis Results

Packaging Text Analysis

Product Information:

Drug Name: PARACETAMOL Paraval-500

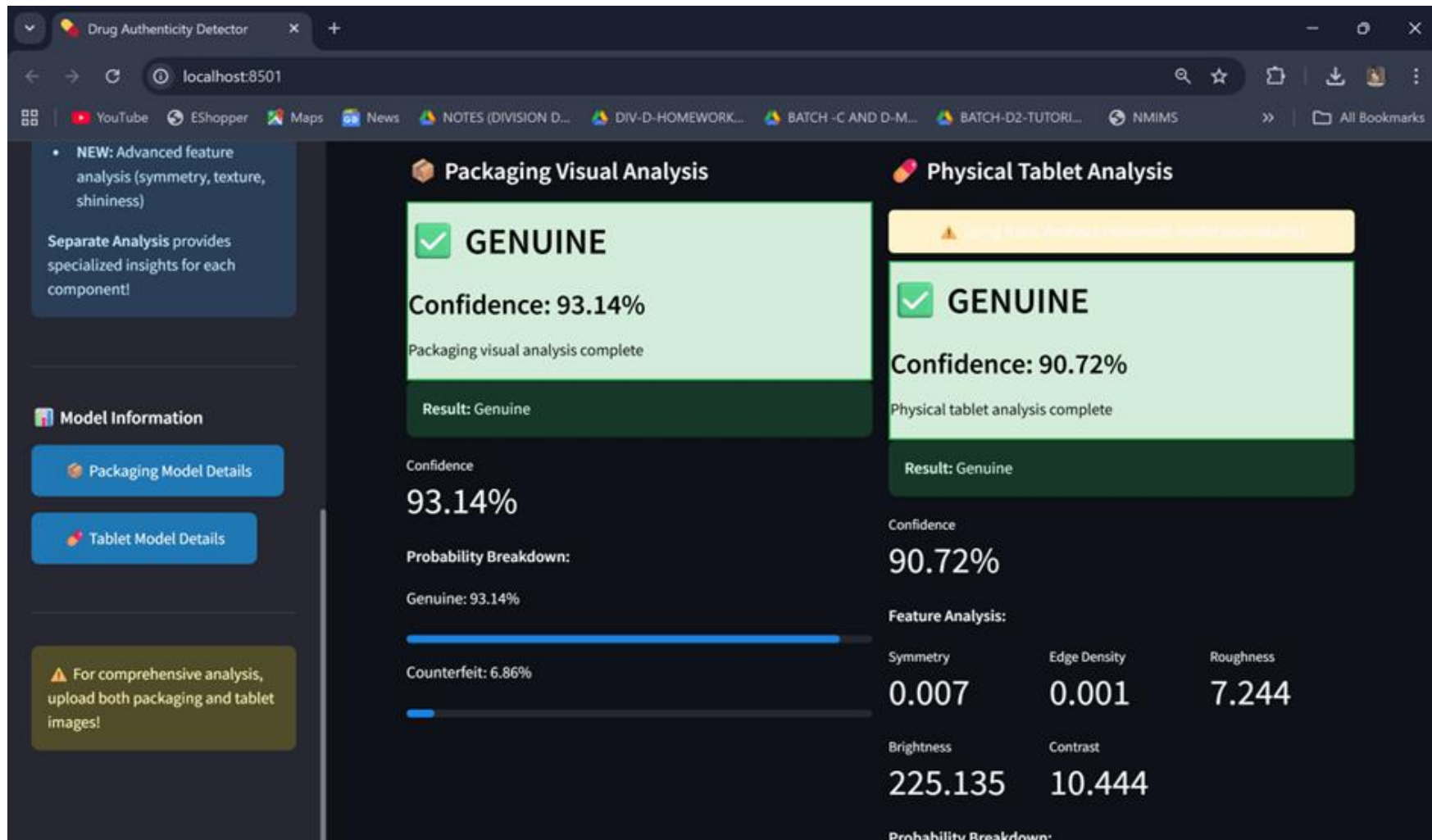
Dosage: 500 MG

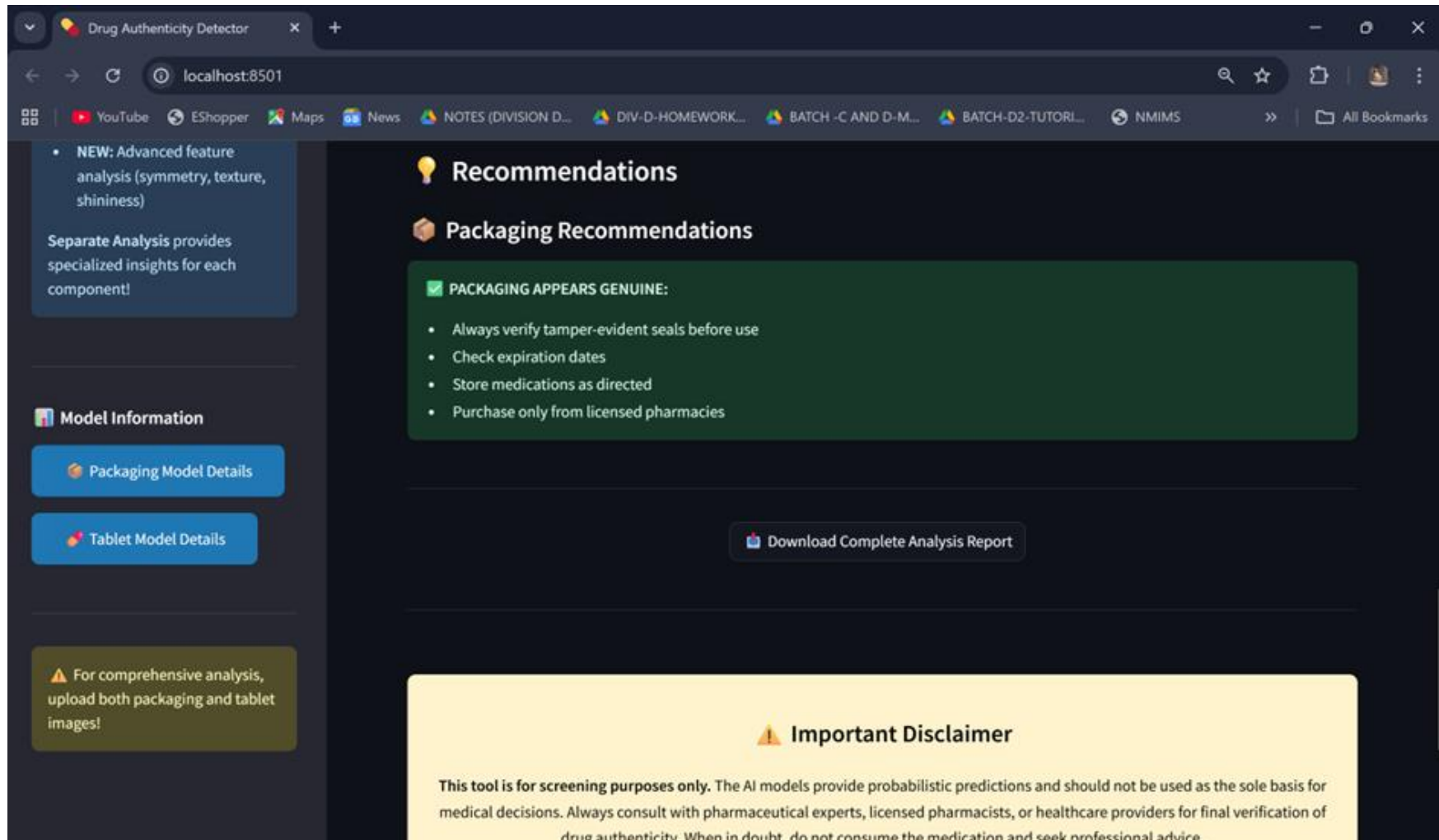
Quantity: TABLETS 500

Packaging Visual Analysis

No tablet image analyzed

GENUINE





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Recommendations

Packaging Recommendations

✅ **PACKAGING APPEARS GENUINE:**

- Always verify tamper-evident seals before use
- Check expiration dates
- Store medications as directed
- Purchase only from licensed pharmacies

Download Complete Analysis Report

Important Disclaimer

This tool is for screening purposes only. The AI models provide probabilistic predictions and should not be used as the sole basis for medical decisions. Always consult with pharmaceutical experts, licensed pharmacists, or healthcare providers for final verification of drug authenticity. When in doubt, do not consume the medication and seek professional advice.

Limitations & Future Enhancements

Current Limitations

Dataset & Scope:

- Limited to solid oral dosage forms (tablets/capsules)
- Geographic bias toward Southeast Asian samples
- Requires controlled lighting for optimal performance

System Constraints:

- Binary classification only (genuine vs counterfeit)
- Cannot detect repackaged authentic medications
- Visual packaging analysis only

Future Roadmap

Near-Term (6-12 months):

- ✓ Expand to liquid formulations & injectables
- ✓ Integrate blockchain for supply chain tracking
- ✓ Multi-language support (currently English-focused)

Long-Term (1-2 years):

- ✓ Explainable AI (Grad-CAM) for transparency
- ✓ Smartphone spectrometry integration
- ✓ GAN-based synthetic counterfeit generation
- ✓ Clinical trials with pharmacy partners

Conclusion & Call to Action

Addressing a \$200 Billion Global Crisis

Our Contribution:

A novel AI-powered multi-modal system that democratizes pharmaceutical authentication through accessible, cost-effective computer vision and deep learning.

Key Differentiators:

- ✓ *Only system addressing real-world class imbalance (11.81:1)*
- ✓ *Perfect counterfeit recall with statistical validation*
- ✓ *Smartphone-accessible for resource-limited settings*
- ✓ *Comprehensive 3-module authentication*

Clinical Readiness:

- *Statistically validated (Cohen's Kappa, MCC, McNemar's test)*
- *Production-ready inference (<2 seconds)*
- *Suitable for regulatory submission*

Next Steps:

1. *Pilot deployment with pharmacy partners*
2. *Collaboration with WHO & regulatory agencies*
3. *Open-source model release for global access*
4. *Clinical validation studies*

Reference

Research Paper Links:

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2. <https://doi.org/10.1016/j.tmaid.2024.102758>
3. <https://www.ncbi.nlm.nih.gov/books/NBK202524/>
4. [http://dx.doi.org/10.13040/IJPSR.0975-8232.3\(11\).4057-64](http://dx.doi.org/10.13040/IJPSR.0975-8232.3(11).4057-64)
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6. <https://doi.org/10.1186/s12911-020-1078-3>
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Datset Reference

Datset Links:

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<https://www.ncbi.nlm.nih.gov/books/NBK202524/>
<https://open.fda.gov/data/downloads/>
<https://www.kaggle.com/datasets/shudhanshusingh/az-medicine-dataset-of-india?resource=download>
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THANK YOU